

The Dual Spacetime Algebra: A Biquaternionic Framework

<https://github.com/hypernumbernet>

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Abstract

The Dual Spacetime Theory (DST), originally motivated by a reformulation of general relativity, reveals a profound algebraic structure isomorphic to the Clifford algebra $\text{Cl}(3, 1)$. This 16-dimensional biquaternionic algebra $\mathbb{H} \otimes \mathbb{H}$ encodes paired Minkowski spacetimes. By associating mathematical objects with rotors in $\text{Spin}^+(3, 1) \oplus \text{Spin}^+(3, 1)$, DST provides an algebraic framework. The invariant, the torsion scalar $J = \frac{1}{16}B(\Omega_{\text{biv}}, \Omega_{\text{biv}})$, derived from the Killing form on the Lie algebra $\mathfrak{so}(3, 1) \oplus \mathfrak{so}(3, 1)$, is explored for its boundedness properties. This paper introduces the algebraic structure and discusses potential mappings to number-theoretic objects, without claiming resolutions of open conjectures.

1 Introduction

The Dual Spacetime Theory (DST) posits that spacetime is not a continuous manifold but a discrete, particle-local structure, each particle carrying a compactified pair of Minkowski spacetimes encoded in the biquaternionic algebra $\mathbb{H} \otimes \mathbb{H} \cong \text{Cl}(3, 1)$ with signature $(-1, +1, +1, +1)$. Primary quaternions $\{i, j, k\}$ and secondary quaternions $\{I, J, K\}$ obey Hamilton's rules ($i^2 = j^2 = k^2 = I^2 = J^2 = K^2 = -1$, $ij = k$, $JI = -K$, cyclic) and commute across copies ($iI = Ii$, etc.), yielding a 16-dimensional real basis: $\{1, i, j, k, I, J, K, iI, iJ, iK, jI, jJ, jK, kI, kJ, kK\}$.

The isomorphism with $\text{Cl}(3, 1)$ identifies:

$$e_0 = j, \quad e_1 = kI, \quad e_2 = kJ, \quad e_3 = kK,$$

satisfying $e_0^2 = -1$, $e_a^2 = +1$ ($a = 1, 2, 3$), $\{e_\mu, e_\nu\} = 0$ ($\mu \neq \nu$). Higher grades follow: bivectors $e_0e_1 = iI$, $e_2e_1 = K$; trivectors $e_1e_2e_3 = k$; pseudoscalar $e_0e_1e_2e_3 = i$.

The usual spacetime vectors are $X = ctj + xkI + ykJ + zkK$, with invariant $X^2 = -(ct)^2 + x^2 + y^2 + z^2$. The dual spacetime is $X' = ct'k + x'jI + y'jJ + z'jK$, with $X'^2 = -(ct')^2 + x'^2 + y'^2 + z'^2$. The dual map $X' = Xi$ reverses time ($ji = -k$) while preserving the norm ($X'^2 = X^2$).

Lorentz transformations arise from rotors $R_{\text{total}} = R_{\text{usual}}R_{\text{dual}}$, where

$$R_{\text{usual}} = \exp\left(\sum_{a=1}^3 \frac{\omega_a}{2} i\Gamma_a\right), \quad R_{\text{dual}} = \exp\left(\sum_{a=1}^3 \frac{\phi_a}{2} \Gamma_a\right),$$

with $\Gamma_1 = I$, $\Gamma_2 = J$, $\Gamma_3 = K$. Boosts ($i\Gamma_a$, $(i\Gamma_a)^2 = +1$) yield hyperbolic expansions ($\cosh(\theta/2) + \sinh(\theta/2)i\Gamma_a$), rotations ($\Gamma_a^2 = -1$) trigonometric ones. The sandwich product $\tilde{X} = R_{\text{total}}X R_{\text{total}}^\dagger$ preserves X^2 .

In DST, curvature is torsion: the mismatch rotor $\Omega = R_{\text{usual}}^\dagger R_{\text{dual}} \in \text{Spin}^+(3, 1)$, bivector $\Omega_{\text{biv}} = \log \Omega \in \mathfrak{so}(3, 1) \oplus \mathfrak{so}(3, 1)$, and scalar

$$J = \frac{1}{16}B(\Omega_{\text{biv}}, \Omega_{\text{biv}}),$$

with Killing form $B(X, Y) = 4 \text{Tr}(XY)$ ($B(i\Gamma_a, i\Gamma_a) = +8$, $B(\Gamma_a, \Gamma_a) = -8$).

This structure generalizes the algebraic encoding to mathematical objects, where each number $n \in \mathbb{R}$ (or $\mathbb{C}$) may be associated with a rotor ensemble via $X = |n| kI$ (spatial projection), with parameters $(\omega_a, \phi_a) \propto \log |n|$. The multiplicity of rotors (6 real parameters) allows flexible mappings. The boundedness of J is explored below as a structural property.

2 The Torsion Scalar and Its Properties

The torsion scalar J is a key invariant in the biquaternionic algebra. While a strict bound $|J| \leq 1$ may hold under additional assumptions on the embedding and chirality (as suggested by the compact nature of the Spin group covering and grade normalizations in $\text{Cl}(3,1)$), we treat the boundedness as a structural hypothesis for exploratory purposes rather than a rigorously established theorem. Further analysis of the Killing form and rotor decompositions is required to confirm any universal bound.

Corollary 2.1 (Rotor Multiplicity). *Each number n admits infinitely many rotors, parametrized by $(\omega_a, \phi_a) \in \mathbb{R}^6$.*

3 Potential Mappings to Diophantine Equations

The rotor association $R(a) = \exp(\log |a| \cdot iI)$ provides a multiplicative homomorphism from positive integers to the boost subgroup. While this does not yield new proofs of known results such as Fermat's Last Theorem (established by Wiles in 1995), it offers an algebraic perspective where additive relations in $\mathbb{Z}$ correspond to non-commutative rotor compositions, potentially illuminating torsion invariants in hypothetical extensions.

4 Analytic Functions and Rotor Sums

The zeta function $\zeta(s)$ may be formally associated with sums of rotors via $n^{-s} \mapsto \exp(-s \log n \cdot iI)$, where the phase shifts reflect the argument of s . This provides an algebraic analogy for studying zero distributions, but does not constitute a proof of the Riemann Hypothesis. Traditional analytic methods remain necessary for such results. The scale-invariance of the algebra may suggest connections to prime distribution heuristics, warranting further investigation.

5 Further Directions

The biquaternionic framework provides a rich algebraic setting for exploring multiplicative and additive structures through rotor compositions and torsion invariants. While direct mappings to Diophantine problems or analytic number theory conjectures do not yield rigorous proofs, the scale-invariant properties and Killing form geometry may inspire new heuristics or computational approaches in number theory. Future work could focus on rigorous definitions of the correspondence and verification of algebraic identities independent of physical interpretations.

6 Conclusion

The Dual Spacetime Algebra, realized through the biquaternionic structure $\mathbb{H} \otimes \mathbb{H} \cong \text{Cl}(3,1)$, offers a novel algebraic perspective on mathematical objects via rotor encodings and torsion invariants. While the framework does not resolve longstanding conjectures in number theory, it highlights interesting connections between Lie algebra geometry and multiplicative structures, inviting further mathematical exploration.

References

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